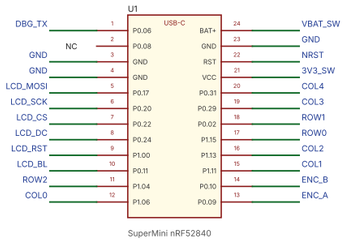
Controller: SuperMini nRF52840 (U1, bottom side, face down)

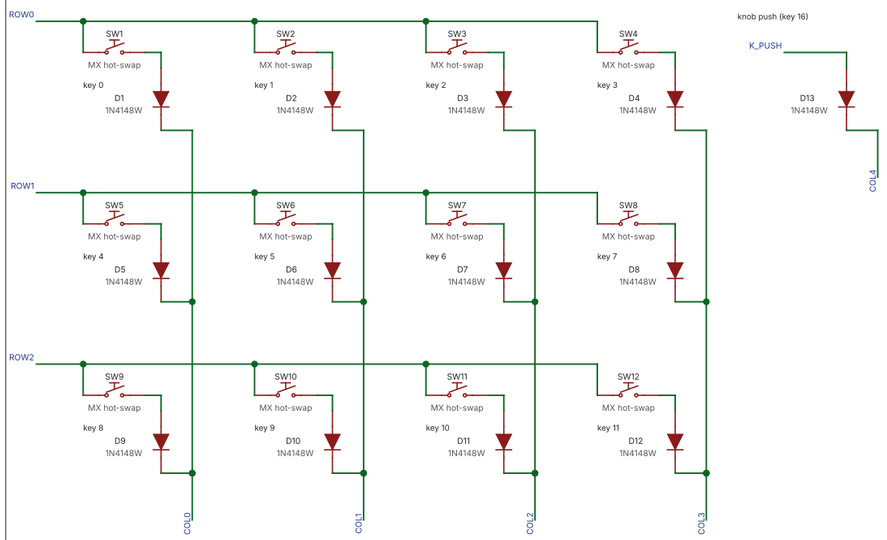


Pin map = firmware/boards/promicro_nrf52840_nrf52840_uf2.overlay

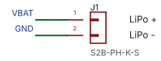
LCD SPI on full-speed pins (P0.20 clock, P0.17 data). P0.09/P0.10 (NFC) carry the knob.

VCC (pin 21) is switched by P0.13: the firmware turns the LCD supply on.

Key matrix: rows = inputs with pull-ups, columns driven low



Power

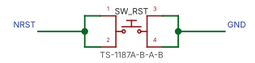


Slide switch in the right wall:

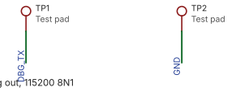
ON = battery to U1 B+ (charger

output; USB charges through it)

Reset, debug, mounting



Double-press = UF2 bootloader



UART log out, 115200 8N1

M2 screws into the case pillars

LCD: Waveshare 2.0in ST7789 module on 8 wires (J2)

Wires in the module's cable order: VCC GND DIN CLK CS DC RST BL

C3 = bulk for the backlight switch-on, next to U1 pin 21

Knob: EC11-size encoder with push (PEC11R-4220F-S0024)

24 detents / 24 pulses: both contacts open at every detent

(no current through the NFC pins at rest). 10 nF + internal pull-up = debounce.

NOTES

1. Every part is in the JR library (JR.SchLib / JR.PcbLib); parameters carry the LCSC numbers.

2. U1 is soldered face down on 2.5 mm low-profile headers. Trim ENC1 pins flush first; Kapton on U1's back.

3. Firmware key n = SW(n+1): row 0 is the back row (next to the screen), SW1 = back-left = top-left tile.

4. Knob push sits in row 0, column 4 with its own diode (D13); firmware key 16 on every concept.

5. Switch sockets, diodes, U1, J1, J2, power and reset are on the bottom; switches and the knob on top.

Macropad concept B main board

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Drawn by pcb/board/schematic.py from design_b.py

github.com/jasoneroberts/macropad